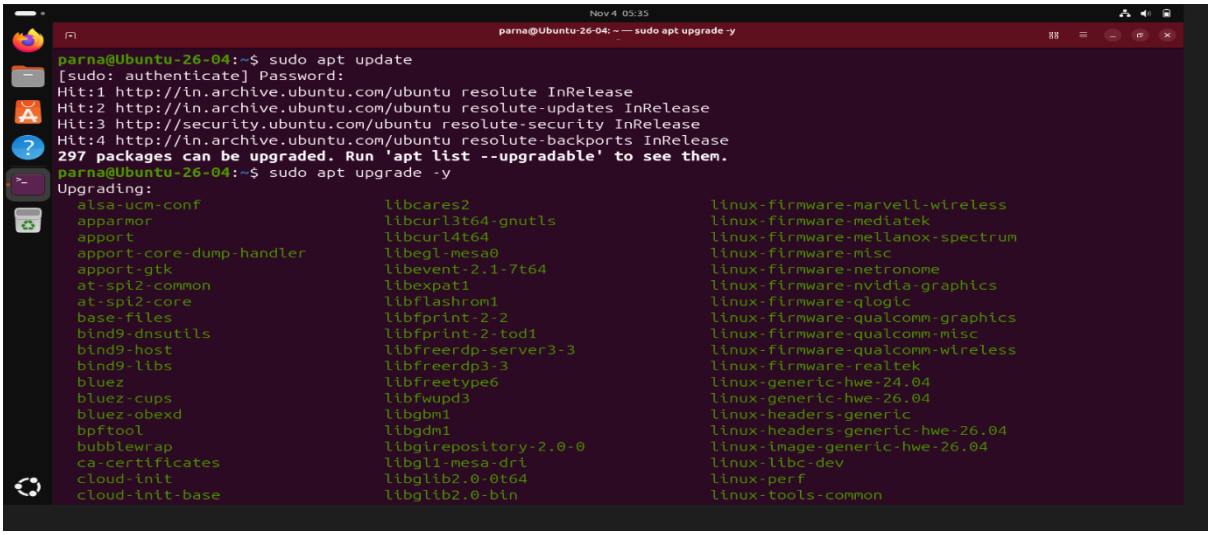
LAB 4: SDN

1. Update Ubuntu

sudo apt update && sudo apt upgrade -y



Updates the Ubuntu system.

2.Check Mininet installation

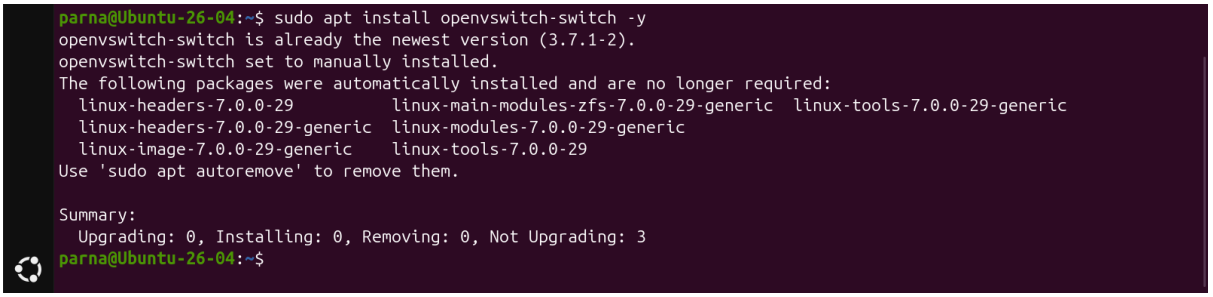
mn --version



Checks Mininet for creating virtual networks.

3. Install Open vSwitch

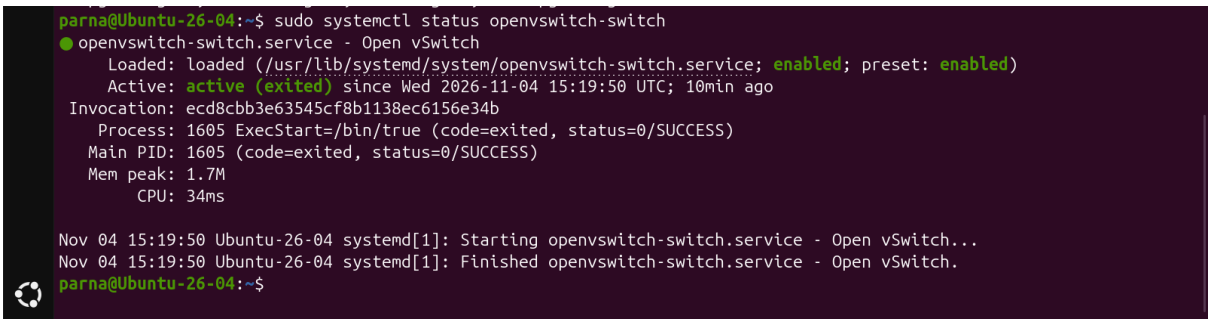
sudo apt install openvswitch-switch -y



Installs the software-based SDN switch.

4. Check Open vSwitch

sudo systemctl status openvswitch-switch



Checks whether Open vSwitch is running.

5. Test Mininet

sudo mn

This starts a basic Mininet network.

mininet> nodes

It displays the available virtual hosts and switch.

mininet> pingall

Tests connectivity between the virtual hosts.

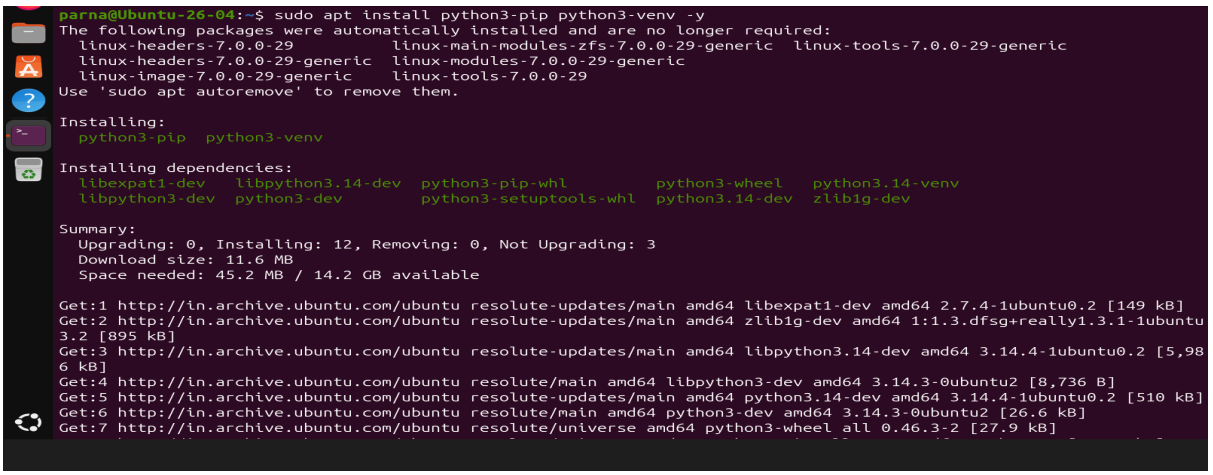
mininet> exit

Exits Mininet.



6. Install Python and Virtual Environment

sudo apt install python3-pip python3-venv -y



Installs Python package and virtual-environment support.

7. Create Ryu Environment

python3 -m venv ryu-env

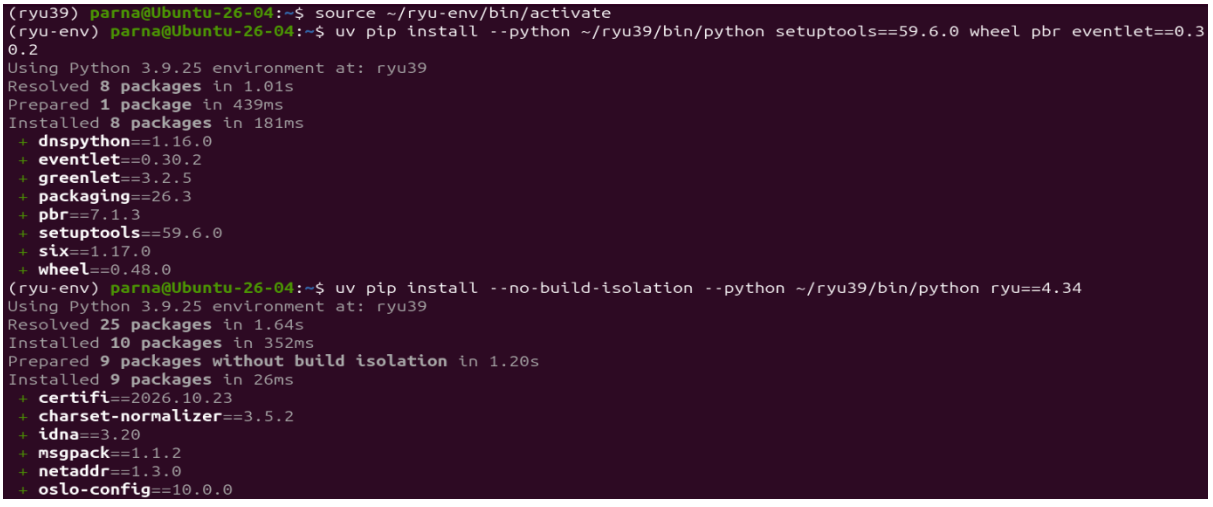
source ryu-env/bin/activate



Creates and activates a Python environment for Ryu.

8. Install Ryu

pip install ryu



Installs the Ryu SDN controller.

9. Test Ryu Installation

ryu-manager –version

```
(ryu-env) parna@Ubuntu-26-04:~$ ~/ryu39/bin/ryu-manager --version
/home/parna/ryu39/lib/python3.9/site-packages/ryu/lib/packet/cfm.py:271: SyntaxWarning: "is not" with a literal. Did
you mean "!="?
  assert interval is not 0
ryu-manager 4.34
```

Displays the installed Ryu version.

10. Start Ryu Controller

ryu-manager ryu.app.simple_switch_13

```
(ryu39) parna@Ubuntu-26-04:~$ ryu-manager ryu.app.simple_switch_13
loading app ryu.app.simple_switch_13
loading app ryu.controller.ofp_handler
instantiating app ryu.app.simple_switch_13 of SimpleSwitch13
instantiating app ryu.controller.ofp_handler of OFPHandler
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:16 2
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:02 2
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:16 1
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:02 1
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:16 2
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:16 3
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:02 3
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:16 1
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:16 3
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:02 2
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:02 3
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:02 1
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:02 3
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:02 1
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:02 2
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:02 3
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:02 1
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:02 2
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:02 3
packet in 1 8e:97:cf:47:2d:c4 33:33:00:00:00:02 1
packet in 1 7a:56:c5:28:39:7f 33:33:00:00:00:02 2
packet in 1 52:c6:29:ea:0f:59 33:33:00:00:00:02 3
```

Starts Ryu with a basic OpenFlow switching application.

Keep this terminal open.

11. Create Mininet SDN Topology

Open a new terminal:

sudo mn --topo single,3 --controller=remote --switch ovsk

```
parna@Ubuntu-26-04:~$ sudo mn --topo single,3 --controller=remote --switch ovsk
[sudo: authenticate] Password:
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
```

Creates 3 virtual hosts connected to an Open vSwitch and remote Ryu controller.

12. Check Nodes

mininet> nodes

```
mininet> nodes
available nodes are:
c0 h1 h2 h3 s1
mininet>
```

Displays H1, H2, H3 and S1.

13. Check Connections

mininet> net

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo:  s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
c0
```

Displays the connections between hosts and the switch.

14. Test Connectivity

mininet> pingall

Verifies communication between all virtual hosts.

mininet> h1 ping h2

Tests connectivity between H1 and H2.

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h1 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.71 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.129 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.128 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.103 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3091ms
rtt min/avg/max/mdev = 0.103/0.516/1.707/0.687 ms
```

15. View Flow Rules

Open another terminal:

sudo ovs-ofctl -O OpenFlow13 dump-flows s1

```
parna@Ubuntu-26-04:~$ sudo ovs-ofctl -O OpenFlow13 dump-flows s1
[sudo: authenticate] Password:
 cookie=0x0, duration=155.979s, table=0, n_packets=9, n_bytes=714, priority=1,in_port="s1-eth2",dl_src=7a:56:c5:28:39
:7f,dl_dst=8e:97:cf:47:2d:c4 actions=output:"s1-eth1"
 cookie=0x0, duration=155.972s, table=0, n_packets=8, n_bytes=616, priority=1,in_port="s1-eth1",dl_src=8e:97:cf:47:2d
:c4,dl_dst=7a:56:c5:28:39:7f actions=output:"s1-eth2"
 cookie=0x0, duration=155.948s, table=0, n_packets=8, n_bytes=672, priority=1,in_port="s1-eth3",dl_src=52:c6:29:ea:0f
:59,dl_dst=8e:97:cf:47:2d:c4 actions=output:"s1-eth1"
 cookie=0x0, duration=155.943s, table=0, n_packets=7, n_bytes=574, priority=1,in_port="s1-eth1",dl_src=8e:97:cf:47:2d
:c4,dl_dst=52:c6:29:ea:0f:59 actions=output:"s1-eth3"
 cookie=0x0, duration=155.908s, table=0, n_packets=3, n_bytes=238, priority=1,in_port="s1-eth3",dl_src=52:c6:29:ea:0f
:59,dl_dst=7a:56:c5:28:39:7f actions=output:"s1-eth2"
 cookie=0x0, duration=155.901s, table=0, n_packets=2, n_bytes=140, priority=1,in_port="s1-eth2",dl_src=7a:56:c5:28:39
:7f,dl_dst=52:c6:29:ea:0f:59 actions=output:"s1-eth3"
 cookie=0x0, duration=328.967s, table=0, n_packets=36, n_bytes=2556, priority=0 actions=CONTROLLER:65535
```

Displays the flow rules installed in the Open vSwitch.

16. Add a Simple Filtering Rule

sudo ovs-ofctl -O OpenFlow13 add-flow s1 "priority=100,in_port=1,actions=drop"

```
parna@Ubuntu-26-04:~$ sudo ovs-ofctl -O OpenFlow13 add-flow s1 "priority=100,in_port=1,actions=drop"
```

Drops packets entering S1 through port 1.

17. Test the Filtering Rule

mininet> h1 ping h2

```
mininet> h1 ping -c 4 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=1.79 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.543 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.094 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.090 ms

--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.090/0.630/1.794/0.696 ms
mininet> h1 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3111ms
```

Checks the effect of the filtering rule on H1's traffic.

18. Remove Flow Rules

sudo ovs-ofctl -O OpenFlow13 del-flows s1

```
parna@Ubuntu-26-04:~$ sudo ovs-ofctl -O OpenFlow13 del-flows s1
```

Removes the manually added flow rules.

19. Clean Mininet

sudo mn -c

```
mininet> h1 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=15.2 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=4.61 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.128 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.120 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3032ms
rtt min/avg/max/mdev = 0.120/5.024/15.234/6.173 ms
mininet> sudo mn -c
```

Cleans up Mininet interfaces and processes.

Result: The SDN environment was successfully set up using Mininet, Open vSwitch, and the Ryu Controller. Basic network traffic forwarding and filtering were also implemented and tested successfully.